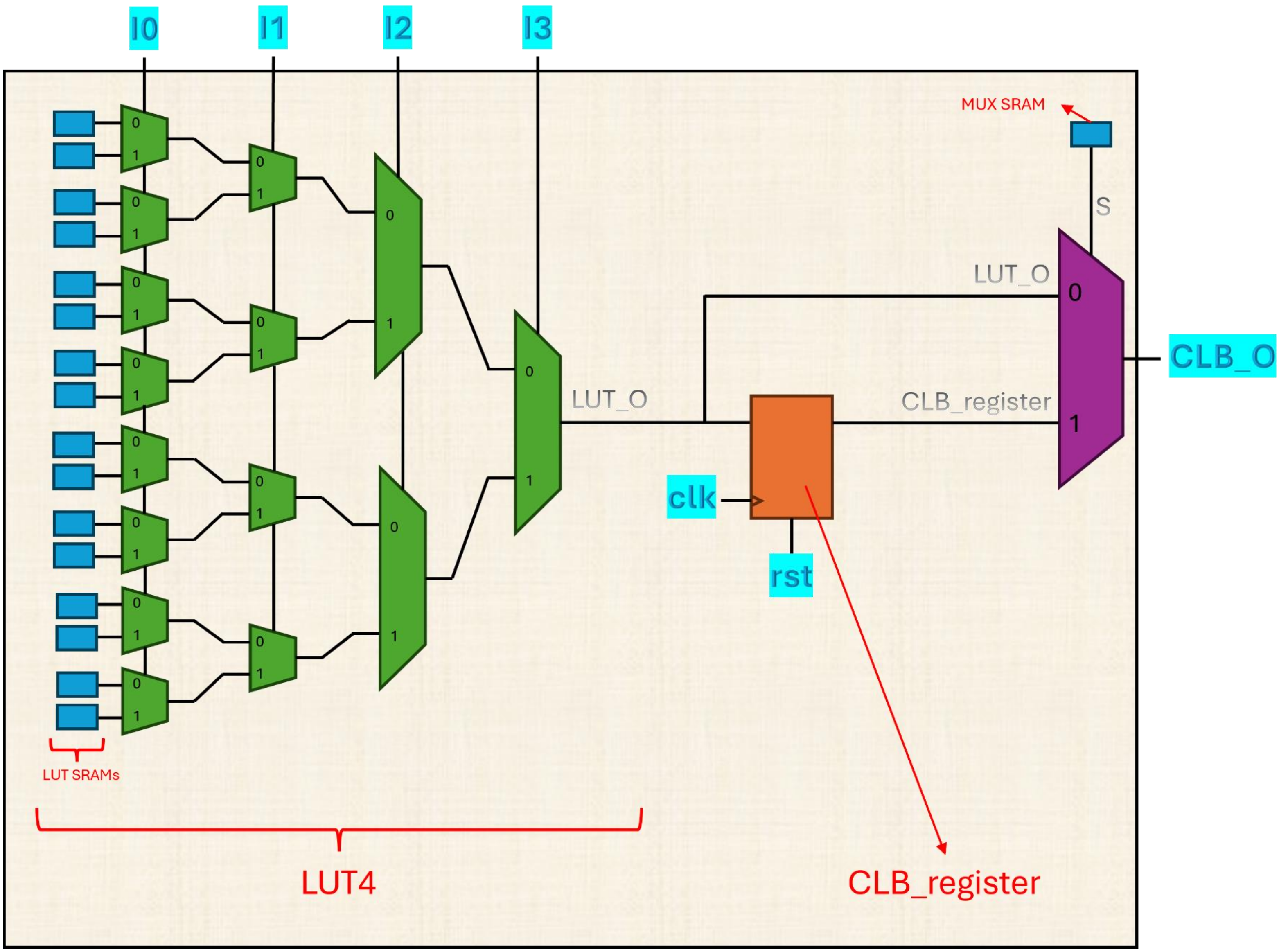
3. In this section, you will create a very simple FPGA and test your design via some combinational and some sequential circuits. You must use MATLAB or Python to verify your design; by creating the exact same program as it is in your HDL.

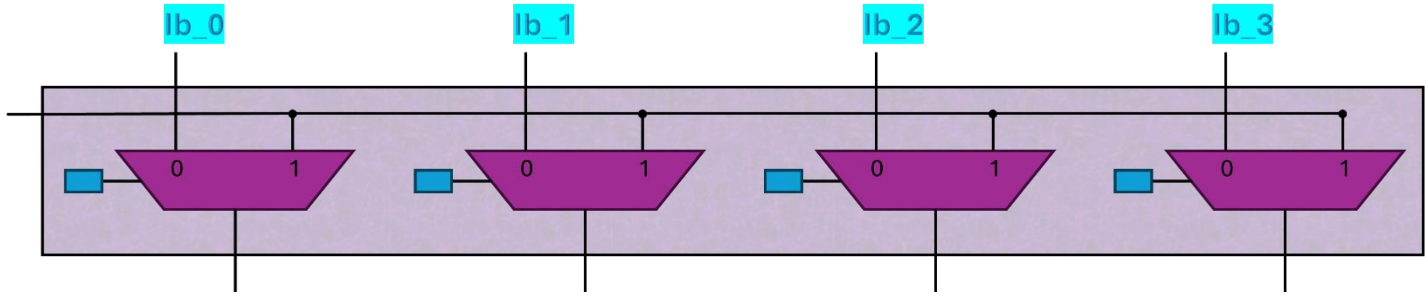
As you can see, a CLB (schematic 1) contains a LUT4 for the combinational section and a flip flop. According to wire “S”, the output of the LUT can either be flopped or not in order to be outputted from the CLB. Each CLB has 17 SRAM cells, 16 for the LUT4 and 1 for the selector of the mux, “S”. Your design must contain 2 CLBs named CLBa and CLBb and 1 connector (schematic 3).

The connector (schematic 2) is to create a versatile connection between the 2 CLBs. It connects the output of CLBa to all the 4 inputs of CLBb via 4 muxes. Each of the muxes has a selector that chooses the input of the CLBb, either it is the output of the CLBa or the input of the FPGA. The selectors within the connector are SRAMs programmed in advance to make sure our desired connections are created.

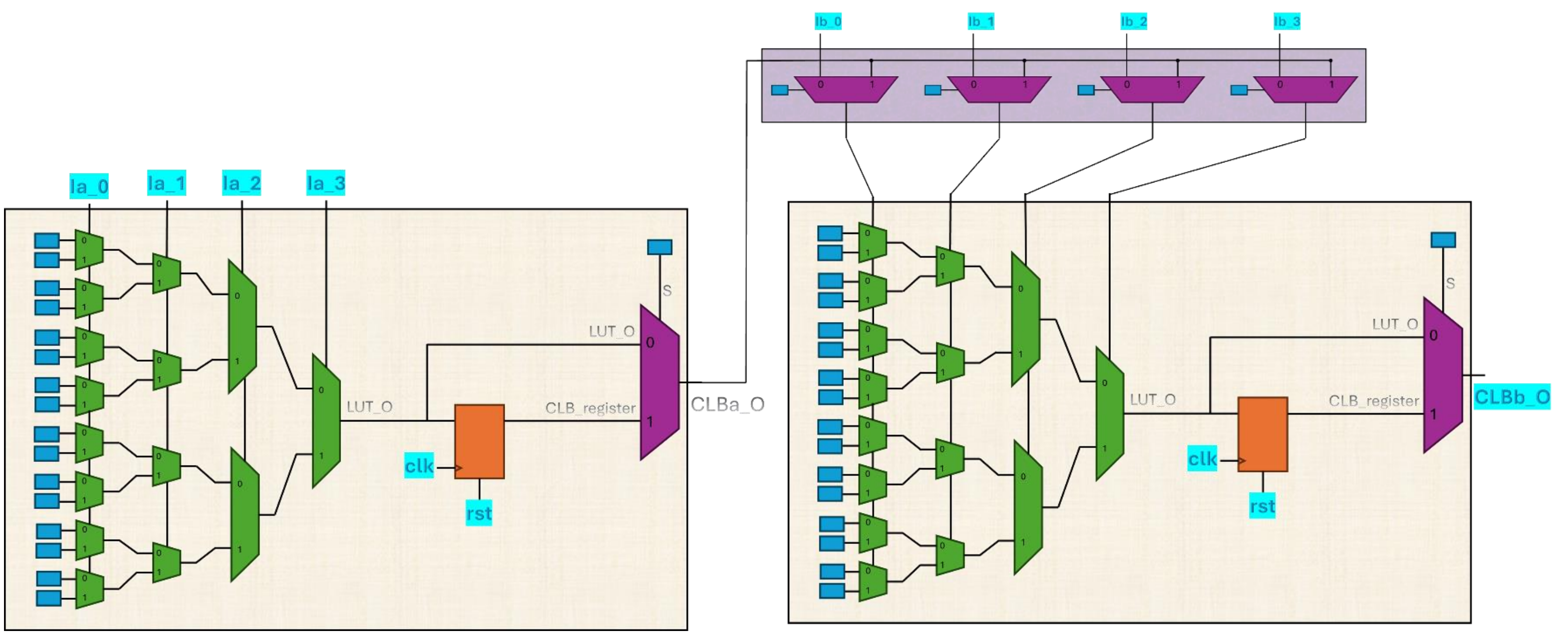
You must have 38 SRAMs in the whole design. You need to create a 38-bit memory for the SRAMs in your design and put the bitstream in the SRAM via your testbench. You may synthesize your design manually to fit your small FPGA and enter the created bitstream into the SRAMs via testbench. Your FPGA module must have 10 inputs, 4 for each of the 2 CLBs in addition to a “clk” and a “rst” for the CLB registers. The FPGA module has only 1 output named “CLBb_O”.



Schematic 1, the “CLB” module



Schematic 2, the “connector” module



Schematic 3, the complete “FPGA” module

You need to implement the following functions using your FPGA. The complete HDL of these functions, as well as the synthesized netlist HDL and the bitstream, must be present in your report.

1) The following truth table (combinational):

a	b	c	d	f
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

2) f = (a&b)|(!c|d) (combinational)
3) f = ((a&b)|(!c|d))^e (output must be registered)
4) f = b>a (a and b are unsigned and 3 bits each and combinational)
5) f = b>a (a and b are unsigned and 3 bits each and pipelined)